UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PRE CALCULUS
COURSE CODE	:	MAT133
EXAMINATION	:	NOVEMBER 2026
TIME	:	2 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the MAT133 mock paper (November 2026).
   Answers are unofficial model answers generated from the printed past-paper question; they are not an official UTM marking scheme.

2. Marking points follow the printed part labels and marks. Verify every model answer against the source page.

3. Four questions: 10 + 10 + 10 + 25 marks; total = 55 marks.

4. Questions: 4. Real recycled questions: 4. Placeholders: 0.
   For local study use only; model answers are unofficial.

5. Every question is a real MAT133 past-paper question; its source paper and page are printed with the answer.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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QUESTION 1 **(10 marks)**

Source: MAT133-2021-JUL, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Solve the inequality $\frac{6}{5-3x} \geq 3$. Express the answer in interval notation.
(5 marks)

Answer: $[1, \frac{5}{3})$

- $\frac{6}{5-3x} - 3 \geq 0 \rightarrow \frac{6-3(5-3x)}{5-3x} \geq 0$
- $\frac{9x-9}{5-3x} \geq 0 \Leftrightarrow \frac{x-1}{5-3x} \geq 0$
- Critical values: $x = 1$ (included), $x = \frac{5}{3}$ (excluded).
- Sign: negative on $(-\infty, 1)$, positive on $(1, \frac{5}{3})$, negative on $(\frac{5}{3}, \infty)$.
- Solution: $[1, \frac{5}{3})$.

b) If $Z_1 = -1 + 2\mathrm{i}$ and $Z_2 = 4 - 3\mathrm{i}$, state $\frac{Z_2}{\overline{Z_1} i^{19}}$ in standard form $a + bi$.
(5 marks)

Answer: $-\frac{11}{5} + \frac{2}{5}i$

- $\overline{Z_1} = -1 - 2i, \qquad i^{19} = i^3 = -i$
- $\overline{Z_1} i^{19} = (-1 - 2i)(-i) = -2 + i$
- $\frac{Z_2}{\overline{Z_1} i^{19}} = \frac{4-3i}{-2+i}$
- $= \frac{(4-3i)(-2-i)}{(-2+i)(-2-i)} = \frac{-11+2i}{5}$
- $-\frac{11}{5} + \frac{2}{5}i$

Model confidence: high.

QUESTION 2 **(10 marks)**

Source: MAT133-2026-FEB, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Solve the inequality $\frac{2x-5}{x-2} \leq 1$ and express the solution in real number line.
(5 marks)

Answer: $x \in (2, 3]$ (equivalently $2 < x \leq 3$); on the number line: open circle at 2, closed circle at 3, and shade the interval between them.

- $\frac{2x-5}{x-2} \leq 1 \Leftrightarrow \frac{2x-5-(x-2)}{x-2} \leq 0$
- $\Leftrightarrow \frac{x-3}{x-2} \leq 0$
- Critical values: $x = 2$ (excluded) and $x = 3$ (included).
- Sign table: $x < 2 : +$; $2 < x < 3 : -$; $x > 3 : +$.
- Thus the solution is $(2, 3]$.

b) Given the equation $2(y-3)^2 + 32 = 16x$, find the standard form of the parabola, the vertex and the equation of directrix. Hence, sketch the parabola with correct vertex and directrix:
(5 marks)

Answer: Standard form: $(y-3)^2 = 8(x-2)$; vertex: $(2, 3)$; directrix: $x = 0$. Sketch: a parabola opening

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to the right with vertex $(2, 3)$ and vertical directrix $x = 0$ (focus $(4, 3)$).

- $2(y-3)^2 + 32 = 16x$
- $2(y-3)^2 = 16x - 32 = 16(x-2)$
- $(y-3)^2 = 8(x-2) = 4(2)(x-2)$

• Compare with $(y-k)^2 = 4p(x-h)$: $h = 2,\ k = 3,\ p = 2$.

• Vertex $(h, k) = (2, 3)$.

• Directrix $x = h - p = 2 - 2 = 0$. Since $p > 0$, the parabola opens right.

Assumptions: No printed grid/figure is provided, so the required sketch is described by its essential features: vertex, directrix, and opening direction.

Model confidence: high.

QUESTION 3 (10 marks)

Source: MAT133-2025-JUL, page(s) 2

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Solve the inequality $(5-x)^2 \le 9$ and express the solution in real number line.
(5 marks)

Answer: $x \in [2, 8]$; on the real number line, place closed circles at $x = 2$ and $x = 8$ and shade the interval between them.

- $(5-x)^2 \le 9$
- $|5 - x| \le 3$
- $-3 \le 5 - x \le 3$
- $-8 \le -x \le -2$
- $2 \le x \le 8$

b) Write the equation of circle $x^2 + y^2 - 4x - 2y - 11 = 0$ in standard form. Hence, sketch the circle with the correct center and radius.
(5 marks)

Answer: Standard form: $(x-2)^2 + (y-1)^2 = 16$. Center: $(2, 1)$; radius: 4. Sketch a circle centered at $(2, 1)$ with radius 4, passing through $(6, 1)$, $(-2, 1)$, $(2, 5)$ and $(2, -3)$.

- $x^2 + y^2 - 4x - 2y - 11 = 0$
- $(x^2 - 4x + 4) + (y^2 - 2y + 1) = 11 + 4 + 1$
- $(x-2)^2 + (y-1)^2 = 16$

• Center $(2, 1)$, radius $r = \sqrt{16} = 4$

Model confidence: high.

QUESTION 4 (25 marks)

Source: MAT133-2025-FEB, page(s) 3

Unofficial model answer — AI-generated, not an official UTM marking scheme.

a) Given that $f(x) = 1 - 3x$ and $g(x) = \sqrt{x+5}$.

i) State the domain of $f(x)$ and $g(x)$.
(2 marks)

ii) Find $(g \circ f)(x)$.
(3 marks)

iii) Find the value of x if $(g \circ f)(x) = 3$.

(4 marks)

Answer: Domain of f: $(-\infty, \infty)$; domain of g: $[-5, \infty)$. | $(g \circ f)(x) = \sqrt{6-3x}$, with $x \leq 2$. | $x = -1$

- $f(x) = 1 - 3x$ is linear, so its domain is all real x.
- $x + 5 \geq 0$ for $\sqrt{x+5}$, so $x \geq -5$.
- $(g \circ f)(x) = g(1-3x) = \sqrt{(1-3x)+5}$.
- Thus $(g \circ f)(x) = \sqrt{6-3x}$.
- Composite domain: $1 - 3x \geq -5 \rightarrow x \leq 2$.
- $\sqrt{6-3x} = 3$.
- $6 - 3x = 9$.
- $-3x = 3 \rightarrow x = -1$.
- Check: $\sqrt{6-3(-1)} = \sqrt{9} = 3$.

b) Using synthetic division, find the value of m if $f(x) = 2x^3 - mx + 10$ has a remainder of 18 when $f(x)$ is divided by $(x-2)$.
(5 marks)

Answer: $m = 4$

- Use synthetic division with root 2 and coefficients $2, 0, -m, 10$.
- Bring down 2; multiply by 2 to get 4; add to 0 to get 4.
- Multiply 4 by 2 to get 8; add to $-m$ to get $8 - m$.
- Multiply $8 - m$ by 2 to get $16 - 2m$; add to 10 to get remainder $26 - 2m$.
- Set $26 - 2m = 18$.
- $2m = 8 \rightarrow m = 4$.

c) Find the value of x if $\log_4(3+x) = 2 + \log_4(x+1)$.
(6 marks)

Answer: $x = -\frac{13}{15}$

- Domain: $x + 3 > 0$ and $x + 1 > 0$, so $x > -1$.
- $\log_4(3+x) - \log_4(x+1) = 2$.
- $\log_4\left(\frac{x+3}{x+1}\right) = 2$.
- $\frac{x+3}{x+1} = 4^2 = 16$.
- $x + 3 = 16x + 16 \rightarrow -13 = 15x$.
- $x = -\frac{13}{15}$, which satisfies $x > -1$.

d) Based on the graph $y = |x|$, sketch the graph for $y = -[|x-2| - 2]$ using transformation technique. Hence, state the y-intercept(s) (if any).
(5 marks)

Answer: $y = -|x-2| + 2$. The graph is an inverted V with vertex $(2, 2)$ and x-intercepts $(0, 0)$ and $(4, 0)$. y-intercept: $(0, 0)$ (i.e. $y = 0$).

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• Start with $y = |x|$.

• Shift right 2: $y = |x - 2|$.

• Shift down 2: $y = |x - 2| - 2$.

• Reflect in the x-axis: $y = -(|x - 2| - 2) = -|x - 2| + 2$.

• Vertex: $(2, 2)$; at $x = 0$, $y = -|-2| + 2 = 0$.

Assumptions: For part (d), the sketch is represented by its transformed equation and key features because the response format is JSON/text.

Model confidence: high.

END OF MARKING SCHEME

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